Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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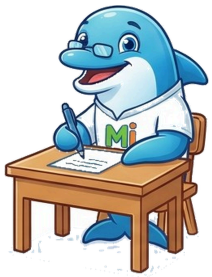
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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2	**Cryptarithm crackers** Not in this proof edition	–
3	**Times table treasure** Times table treasure · follows Holiday Program Lesson 3 · answers on page 16	page 6
4	**Model makers (bar models)** Not in this proof edition	–
5	**Remainder riddles** Not in this proof edition	–
6	**Fraction walls** Not in this proof edition	–
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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	Not in this proof edition				☐
3	Times table treasure	6-7	8-9	10 and 12	11	☐
4	Model makers (bar models)	Not in this proof edition				☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	Not in this proof edition				☐
7	Area architects	Not in this proof edition				☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

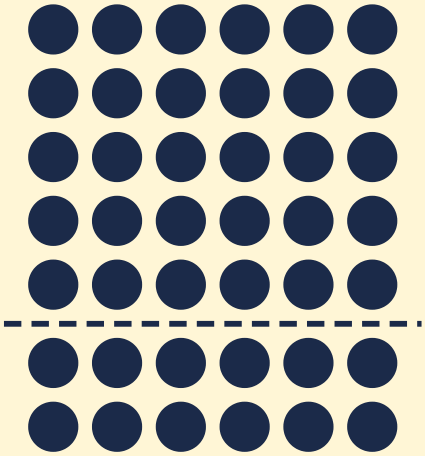
Learn it: Tables of 6, 7, 8 and 9

My goal. I can use facts I know to find new times table facts. I can find patterns and solve number puzzles.

A times table fact you do not know can be built from facts you do know. Split a rectangle of dots into two smaller rectangles.

Worked example

There are 7 rows with 6 dots in each row. How many dots are there in all?



1. The dashed line splits the dots into 5 rows and 2 rows.
2. 5 × 6 = 30 and 2 × 6 = 12.
3. Add the 2 parts: 30 + 12 = 42.
4. So 7 × 6 = 42.

Answer: 7 × 6 = 42. There are 42 dots.

Ways to find a fact

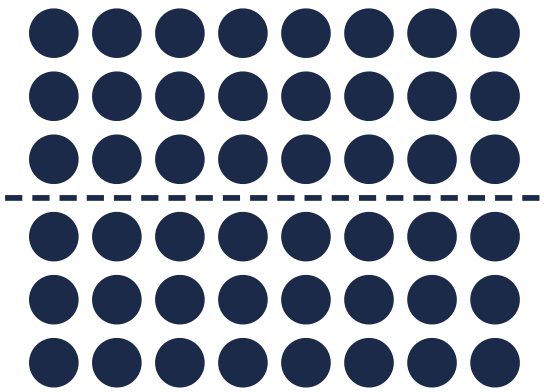
1. Split into facts you know: 5 ×, 2 ×, 10 ×.
2. For 9 ×: take 1 group away from 10 groups.
3. Check with the last digit.

Your turn

Answer at the back of the book.

Try

There are 6 rows with 8 dots in each row. Split them into 3 rows and 3 rows. How many dots are there in all?



Start here

Do these first. Take your time.

1 Write the missing answers in the 8 times table.

n	1	2	3	4	5	6	7	8	9	10
8 × n	8	16	24		40		56	64		80

2 Write the answers.

(a) 6 × 8 = ☐ (b) 7 × 9 = ☐ (c) 9 × 6 = ☐

(d) 8 × 7 = ☐

3 You know that 8 × 5 = 40. Use it to find the other facts.

(a) 8 × 6 = ☐

(b) 8 × 4 = ☐

(c) 8 × 10 = ☐

(d) 8 × 9 = ☐

4 Ali packs 7 eggs in each box. He fills 9 boxes.

(a) He packs ☐ eggs.

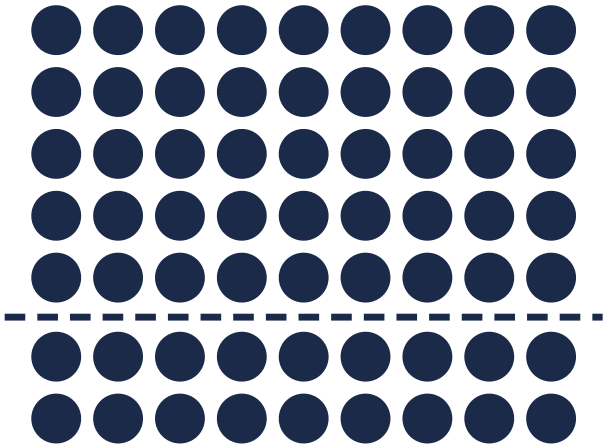
(b) Then he packs 56 eggs into boxes of 7. He fills ☐ boxes.

5 Find my number. It is in the 7 times table, and it is greater than 50 and less than 80. Its digits add up to 9. ☐

Start here

Keep going. Check your work.

6 **The dashed line splits 7 rows of 9 dots into 2 parts.**



(a) The top part has 5 × 9 = ☐ dots.

(b) The bottom part has 2 × 9 = ☐ dots.

(c) There are ☐ dots in all.

7 **Fill in the missing answers.**

×	6	7	8	9
6	36	42	A	54
7	B	49	56	C
8	48	D	64	72

A = ☐ B = ☐ C = ☐ D = ☐

8 **A row number times a column number makes each product. Write the missing numbers in the heading cells.**

×		
	54	63
	48	56

Start here

Nearly there. You can do it.

9 **Write the last digit of each answer. Do not find the whole answer.**

(a) 6 × 13 ☐

(b) 8 × 27 ☐

(c) 9 × 45 ☐

(d) 7 × 68 ☐

10 **Mei puts 30 tiles in equal rows. There are at least 2 rows and at least 2 tiles in each row. List ALL the ways.**

There are ☐ ways.

rows	tiles in each row

11 **Which of these numbers are in the 6 times table?**
24, 38, 42, 56, 60

_____

12 **Mei buys 6 packs of 8 stickers. Ben buys 7 packs of 7 stickers.**

(a) Who has more stickers? Mei Ben

(b) How many more? ☐

Think harder

Hints are upside down. Try first!

1 **Ben writes the 7 times table from 7 × 1 to 7 × 30. How many of his answers end in 4?** ☐

The 7 times table: _____

_____

The answers ending in 4: _____

_____

Hint. Which number times 7 ends in 4? Then find the next ones.

2 **The row numbers and column numbers are all from 6 to 9. Find the empty cells A to D. You can write in the heading cells to help you.**

(a) A = ☐ (b) B = ☐

(c) C = ☐ (d) D = ☐

×			
	42	48	A
	49	B	63
	C	D	72

Hint. Start with 49. Which row and column make 49?

3 **Two numbers have a product of 63 and a sum of 16. What are the 2 numbers?** ☐ **and** ☐

Pairs that multiply to 63: _____

_____

_____

Hint. List the pairs of numbers that multiply to make 63.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Digit sums and the 9s.**
Add the digits again and again: 684 gives 6 + 8 + 4 = 18, then 1 + 8 = 9.
Numbers in the 9 times table always end at 9. Which of these are in it?
738, 825, 963, 1089

_____

Hint. Add the digits of each number. Keep adding until 1 digit is left.

2 **Beyond Level 3: Adding every row.**
A hall has 8 rows. Row 1 has 7 seats. Each row has 2 more seats than the row before.

Row	1	2	3	4	5	6	7	8
Seats	7	9						

(a) Row 8 has ☐ seats.

(b) The hall has ☐ seats in all.

Hint. Pair the first and last rows. Then the next pair.

3 **Make your own 2 by 2 grid puzzle with numbers from 6 to 9. Show 3 products. Only ONE answer may fit. Swap it with a grown-up.**

_____

Hint. Pick the row and column numbers first. Then multiply.

4 **Ben multiplies a number by 8.**

(a) His answer ends in 6. The last digit of his number could be ☐ or ☐.

(b) His answer ends in 5. The last digit of his number could be

0 2 5 7 none .

Hint. List the last digits of 8 × 1 up to 8 × 10.

✔ Check yourself

No hints. Try each one on your own.

1 A box holds 8 pens. How many pens are in 7 boxes? ☐

Working: _____

2 You know that 5 × 7 = 35. Use it to find 6 × 7. ☐

Working: _____

3 What is the last digit of 7 × 56? ☐

Working: _____

4 Mei has 54 apples. She puts 6 apples in each bag. How many bags does she need? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 3 · Times table treasure

Your turn
page 6

Answer: 48
Why: 3 × 8 = 24 and 3 × 8 = 24. Then 24 + 24 = 48. So 6 × 8 = 48.

START HERE

1
page 7

Answer: 32, 48, 72
Why: The answers go up by 8 each time. 8 × 4 = 32, 8 × 6 = 48 and 8 × 9 = 72.

2
page 7

Answer: (a) 48 (b) 63 (c) 54 (d) 56
Why: 6 × 8 = 48, 7 × 9 = 63, 9 × 6 = 54 and 8 × 7 = 56. Swap the numbers to check: 8 × 6 = 48.

3
page 7

Answer: (a) 48 (b) 32 (c) 80 (d) 72
Why: (a) One more 8: 40 + 8 = 48. (b) One less 8: 40 - 8 = 32. (c) Double 8 × 5: 80. (d) 8 × 10 = 80, take away one 8: 72.

4
page 7

Answer: (a) 63 (b) 8
Why: (a) 9 × 7 = 63. (b) 56 ÷ 7 = 8, because 7 × 8 = 56.

5
page 7

Answer: 63
Why: The numbers in the 7 times table between 50 and 80 are 56, 63, 70 and 77. Their digits add up to 11, 9, 7 and 14. Only 63 has digits that add up to 9.

6
page 8

Answer: (a) 45 (b) 18 (c) 63
Why: 5 × 9 = 45 and 2 × 9 = 18. Add: 45 + 18 = 63. So 7 × 9 = 63.

7
page 8

Answer: A = 48, B = 42, C = 63, D = 56
Why: Each cell is the row number × the column number. A = 6 × 8 = 48. B = 7 × 6 = 42. C = 7 × 9 = 63. D = 8 × 7 = 56.

8
page 8

Answer: Columns 6 and 7. Rows 9 and 8.
Why: 54 and 63 are in the same row. Both are in the 9 times table: 54 = 9 × 6 and 63 = 9 × 7. So the row number is 9 and the columns are 6 and 7. The second row: 48 = 8 × 6 and 56 = 8 × 7, so the row number is 8.

9
page 9

Answer: (a) 8 (b) 6 (c) 5 (d) 6
Why: Only the last digits matter. (a) 6 × 3 = 18 ends in 8. (b) 8 × 7 = 56 ends in 6. (c) 9 × 5 = 45 ends in 5. (d) 7 × 8 = 56 ends in 6.

10
page 9

Answer: 6
Why: Check the number of rows from 2 upwards: 2 rows of 15, 3 rows of 10, 5 rows of 6, 6 rows of 5, 10 rows of 3 and 15 rows of 2. 4 rows does not work because 30 is not in the 4 times table. That is 6 ways.

11
page 9

Answer: 24, 42, 60
Why: 6 × 4 = 24, 6 × 7 = 42 and 6 × 10 = 60. The numbers 38 and 56 are not: 6 × 6 = 36 and 6 × 9 = 54, and the next answer after 54 is 60.

12 page 9
Answer: (a) Ben (b) 1
Why: Mei has 6 × 8 = 48 stickers. Ben has 7 × 7 = 49 stickers. Ben has 49 - 48 = 1 more.

THINK HARDER

1 page 10
Answer: 3
Why: 7 × 2 = 14 ends in 4. The last digits of the 7 times table repeat every 10 numbers, so the next ones are 7 × 12 = 84 and 7 × 22 = 154. 7 × 32 is beyond 7 × 30. That makes 3 answers.

2 page 10
Answer: (a) 54 (b) 56 (c) 56 (d) 64
Why: 49 = 7 × 7, so row 2 and column 1 are both 7. Then 42 = 6 × 7 gives row 1 = 6, and 48 = 6 × 8 gives column 2 = 8. 63 = 7 × 9 gives column 3 = 9. 72 = 8 × 9 gives row 3 = 8. A = 6 × 9 = 54, B = 7 × 8 = 56, C = 8 × 7 = 56 and D = 8 × 8 = 64.

3 page 10
Answer: 7 and 9
Why: The pairs that multiply to 63 are 1 and 63, 3 and 21, 7 and 9. Their sums are 64, 24 and 16. Only 7 and 9 add up to 16.

GO FURTHER

1 page 11
Answer: 738, 963, 1089
Why: 738: 7 + 3 + 8 = 18, then 1 + 8 = 9. 825: 8 + 2 + 5 = 15, then 1 + 5 = 6, so not. 963: 9 + 6 + 3 = 18, then 9. 1089: 1 + 0 + 8 + 9 = 18, then 9. Check: 738 = 9 × 82, 963 = 9 × 107, 1089 = 9 × 121.

2 page 11
Answer: (a) 21 (b) 112
Why: The seats go 7, 9, 11, 13, 15, 17, 19, 21. Row 8 has 21 seats. Pair them: 7 + 21 = 28, 9 + 19 = 28, 11 + 17 = 28, 13 + 15 = 28. There are 4 pairs, so 4 × 28 = 112 seats.

3 page 11
Answer: Answers vary. Example: the products 42, 48 and 49 are shown. The row numbers are 6 and 7 and the column numbers are 7 and 8, so the hidden product is 56.
Why: Check that each shown product belongs to the numbers you chose. Check that no other row and column numbers from 6 to 9 give the same 3 products. In the example, 42 = 6 × 7 and 48 = 6 × 8 give row 6 and columns 7 and 8. Also 49 = 7 × 7 gives row 7. The hidden product is 7 × 8 = 56.

4 page 11
Answer: (a) 2 or 7 (b) none
Why: The last digits of the 8 times table are 8, 6, 4, 2, 0, 8, 6, 4, 2, 0. A last digit of 6 comes from 8 × 2 and 8 × 7. A last digit of 5 never comes: every answer in the 8 times table is even.

CHECK YOURSELF

1 page 12
Answer: 56
Why: 7 × 8 = 56. Check: 8 × 7 = 56.

2 page 12	**Answer:** 42 **Why:** 6 × 7 is one more 7 than 5 × 7: 35 + 7 = 42.
3 page 12	**Answer:** 2 **Why:** Only 7 × 6 matters for the last digit. 7 × 6 = 42 ends in 2. (7 × 56 = 392.)
4 page 12	**Answer:** 9 **Why:** 54 ÷ 6 = 9, because 6 × 9 = 54.